

# H-Phase: OntoSage Evaluation Summary

N = 7,151 questions drawn from the pre-development survey corpus (81 participants, Phases A-G classified)

## Headline Metrics

| Metric                                 | Value   | 95% CI       |
|----------------------------------------|---------|--------------|
| Answer rate (Grounded + Informational) | 66.1%   | [65.5-67.6%] |
| Data-grounded rate                     | 26.0%   | [25.0-27.0%] |
| Disambiguation rate                    | 25.5%   | --           |
| Graceful refusal rate                  | 7.1%    | --           |
| Timeout/failure rate                   | 0.9%    | --           |
| Median latency                         | 4.90 s  | --           |
| P90 latency                            | 14.45 s | --           |

## Outcome Definitions

- \* GROUNDED -- response includes specific building data (sensor readings, tables, measured values)
- \* INFORMATIONAL -- general domain knowledge without specific building data
- \* DISAMBIGUATION -- system requested clarification with numbered options
- \* BOUNDARY -- graceful refusal: data not available or out of scope
- \* FAILED -- timeout (60 s limit) or empty response

## Output Files

| File                             | Description                             |
|----------------------------------|-----------------------------------------|
| H1_overall_metrics.csv           | Overall outcome counts and rates        |
| H2_answer_rate_by_domain.csv     | Domain-level answer rates with CI       |
| H3_answer_rate_by_query_type.csv | Query-type answer rates                 |
| H4_answer_rate_by_complexity.csv | L1/L2/L3 complexity analysis            |
| H5_answer_rate_by_stage.csv      | Stage S1-S4 answer rates                |
| H6_latency_by_outcome.csv        | Latency statistics by outcome           |
| H7_domain_complexity_heatmap.csv | Domain x Complexity matrix              |
| H8_domain_stage_heatmap.csv      | Domain x Stage matrix                   |
| H9_answer_rate_by_persona.csv    | Persona-based answer rates              |
| H10_statistical_tests.csv        | All statistical tests with effect sizes |
| H11_answer_rate_by_intent.csv    | Intent-type answer rates                |

| Figure                       | Description                   |
|------------------------------|-------------------------------|
| H1_outcome_distribution      | Stacked proportion bar        |
| H2_answer_rate_by_domain     | Grouped horizontal bar        |
| H3_outcome_by_query_type     | Stacked bar by query type     |
| H4_outcome_by_complexity     | Stacked bar + latency overlay |
| H5_answer_rate_by_stage      | Line plot with CI ribbon      |
| H6_latency_analysis          | Box plot + CDF                |
| H7_domain_complexity_heatmap | Heatmap with annotations      |
| H8_domain_stage_heatmap      | Heatmap with annotations      |
| H9_answer_rate_by_persona    | Horizontal bar with CI        |
| H12_coverage_bubble          | Coverage scatter              |